# AgenticRotation: Learning Local Control for Asynchronous Visual Agents in Semantic In-Hand Reorientation

Anonymous CVPR submission

## Abstract

*A visual agent can decide which face of an object a user wants to see, but a dexterous hand must maintain contact while that decision is being computed. We study semantic in-hand reorientation with asynchronous visual reasoning and continuous physical execution. AgenticRotation combines a frozen RGB pose estimator, a recurrent student conditioned on a semantic goal representation, and lightweight controllers learned for pending requests and subsequent execution. The controllers use current measurements, the active goal, and request history to modulate the student's joint targets. Privileged teacher actions supervise training without entering the deployed policy. A frozen protocol evaluates the requested semantic binding and true orientation at autonomous stopping, separately from physical reach. On one marked cube with 40 raw trials per system, the full Astra system succeeds in 29 trials, a fine-tuned VLM without the added controllers in 27, and a rule-based VLM in 14. Under the common service-eligible set, the rates are 74.4%, 66.7%, and 33.3%. The small difference from the fine-tuned VLM is not statistically resolved. Phase-level analysis identifies unbound inspection as the main remaining failure source.* ***[TODO: Complete the additional-object, predictor, and real-robot studies before submission.]***

## 1. Introduction

A command such as "turn the face marked T upward" specifies a visual relation, not a sequence of joint targets. A robot must locate the mark, associate it with a controllable orientation, and move the object without losing it. The requested face may initially be hidden by the object itself or by the fingers. Moving to obtain evidence changes both visibility and contact. This coupling makes in-hand reorientation a useful setting for studying visual agents that act through physical feedback.

Large visual models provide a flexible interface between an instruction and an action. Their inference time introduces another problem. The hand cannot assume that the object remains in place while an external model reasons. Continuing the current skill can expose useful evidence or destabilize the grasp. Slowing every action can reduce movement without supplying the corrective motion needed to recover. A suitable local controller must act on the current contact state even when the next semantic decision is unavailable.

We investigate the following question: *how should a dexterous controller execute a visual agent's current intention while a new decision is pending, and resume useful motion after it arrives?* Our system separates the semantic decision rate from the motor control rate. An asynchronous agent requests inspection, binds a visible target, and updates its command from feedback. A recurrent student maps a goal representation and measured object pose to joint targets. Two compact learned controllers adapt those targets during pending requests and ordinary execution. This design permits a local policy to react without a second expensive reasoning call.

The visual and control interfaces are explicit. We adapt a ViserDex-style RGB-to-pose policy [1] to a continuous semantic goal input while reusing compatible recurrent weights. A fine-tuned local VLM directly predicts a goal vector. The Astra implementation instead emits validated public tool arguments that are mapped to the same goal space. Thus the controller consumes a common representation, but the Astra branch does not expose hidden model embeddings. The distinction makes it possible to compare a strong external agent with a compact learned grounding model without changing the deployed motor interface.

A credible evaluation must distinguish seeing the target, moving toward it, and finishing correctly. We therefore freeze the task mapping, checkpoint identities, stopping path, and physical success rule before the reported evaluation. Each system retains 40 raw trials. Service-related exclusions are applied symmetrically within a comparison and reported alongside the raw counts. Physical reach diagnostics use a separate metric. This avoids attributing a high low-level reach rate to successful semantic reasoning or autonomous stopping.

The present evidence supports a working semantic reorientation system and a specific control-learning hypothesis. On the frozen cube release, learned waiting and execution produce 29 successes compared with 23 under fixed slowing and the unmodified student. The fine-tuned VLM is com-

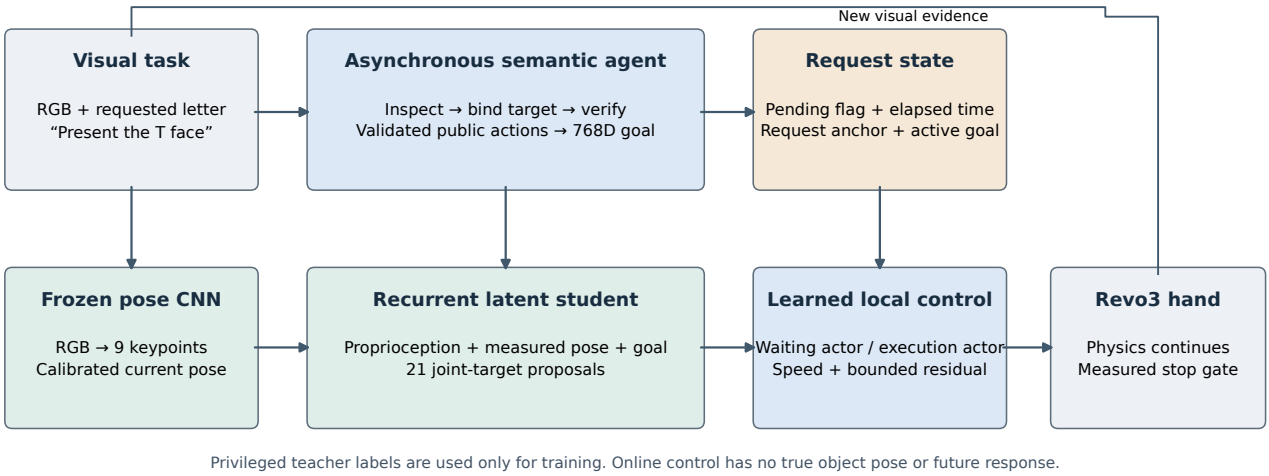



Figure 1. **Continuous control around asynchronous semantic decisions.** The visual agent selects an inspection or target skill. Its validated public command determines a semantic goal vector. A frozen RGB pose estimator and recurrent student run underneath this interface. Separate learned actors adapt joint targets while a request is pending and during execution. The hand continues moving during inference. A common measured-pose gate decides when to stop, and an independent evaluator checks true task completion. The fine-tuned VLM baseline predicts its vector directly and uses the same student without the added actors.

petitive with Astra. Current sample sizes do not establish an advantage over that baseline or a reliable interaction between the two controllers. We report these limits because they determine which experiments can substantiate the proposed method.

This draft develops three contributions: (1) a pose-mediated semantic control interface that connects asynchronous visual agents to a recurrent dexterous student, (2) phase-specific control learning from privileged teacher targets using deployable observations and active goals, and (3) a matched evaluation that separates semantic completion, contact failure, and service availability. The first cube study is complete. Multi-object transfer, causal tests of semantic conditioning, and hardware evaluation remain open experimental requirements.

## 2. Related Work

**Visual dexterous manipulation.** ViserDex uses monocular visual pose estimation and recurrent control for dexterous reorientation, with Gaussian representations supporting visual training [1]. We reuse this division between perception and control and adapt a released Revo3 policy to semantic goals. ManipTrans studies residual learning for dexterous bimanual transfer [4]. These works motivate a reusable motor policy and residual adaptation. Our residual parameterization alone is not a novelty claim. The question here is how a local policy should condition its behavior on the phase and intention of an asynchronous semantic agent.

**Language and visual agents.** Magma connects multimodal representations to agent actions [10]. VERA translates visual plans into robot actions and includes language-conditioned dexterous in-hand reorientation [5]. Its hand experiments include directional rotation prompts. Consequently, language-guided in-hand rotation is not an unoccupied task category. Our setting emphasizes identifying a requested marked face under occlusion, maintaining contact during external reasoning, and checking a canonical target at autonomous stopping. Those distinctions require direct evidence, rather than a claim of being the first language-controlled hand.

**Asynchronous robot policies.** Real-time chunk execution [2], online corrections for VLA chunks [8], and latency-aware RL [11] address inference delays through different execution and learning designs. DEFLECT studies delay-robust asynchronous VLAs through temporal counterfactual preferences [12]. LiMA combines sparse semantic intention and asynchronous action generation for dexterous manipulation [3]. These are close methodological references. Our current implementation learns small teacher-supervised controllers around a separate recurrent policy and public semantic tools. A controlled comparison with a generic time-aware residual policy is still needed to show that the active semantic goal and request anchor add value beyond reactive correction.

## 3. Task and Evaluation Contract

**Semantic target.** The current object is a cube with six marked faces. An instruction identifies one letter from $\{T, E, R, L, S, H\}$. Each face has a fixed color and a canonical world orientation $Q(g)$ that places its normal upward. The environment scores the full canonical quaternion, including its fixed twist, rather than accepting every orientation with the correct normal. The language prompt is therefore less specific than the scoring convention. This convention is disclosed and held constant for all methods. A face-normal-only score can be reported as an additional diagnostic without replacing the primary metric.

At step $t$, the agent observes RGB evidence $I_t$ and the instruction $\ell$. The motor system observes proprioception $x_t$ and measured pose $\hat{p}_t$. True pose $p_t$, teacher actions, and future semantic responses are unavailable to online control. The simulator exposes them only for training labels and independent scoring. We denote the active goal vector by $g_t$, recurrent state by $h_t$, and the pending-request flag by $m_t$.

**Physical time.** One request may be pending at a time. Each episode allows six model calls and 3,600 logical control steps at 30 Hz. If a call starts at $k$ and takes $L$ wall-clock seconds, its response becomes usable only after the call completes and at least $\lceil 30L \rceil$ control steps have elapsed. Physics continues throughout this interval. The simulator currently runs slower than real time. This protocol represents inference delay in simulated physical time, but does not demonstrate a 30 Hz hardware pipeline.

**Autonomous completion.** All systems share a measured-pose stopping gate, evaluated at 3 Hz. It requires a declared binding for the requested symbol, no pending request, a non-inspection state, and measured full-orientation error at most $\epsilon = 0.16$ radians. The gate may stop independently of an agent's native stop suggestion. At stop time $\tau$, the independent success variable is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared face binding, $q_\tau$ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. Merely passing through a target does not count. No extra hold duration is added after stopping. A pose false stop and a semantically wrong binding are failures even if the hand retains the object. The frozen drop rule is distance of at least 0.24 metres from the nominal object grasp position.

**Matched reporting.** Every arm uses 40 raw trials from two seeds, with 20 trials per seed and matching initial physical states and instructions across semantic conditions. A service error or measured model latency above 120 seconds is recorded as a service exclusion. For a panel of methods $\mathcal{C}$, the eligible set removes the union of excluded trial identifiers across $\mathcal{C}$. We report both raw and eligible outcomes, never replace an excluded trial, and never mix denominators from different panels. Long service delays remain visible without being interpreted as controller failures.

## 4. Method

Figure 1 summarizes AgenticRotation. The system has a semantic interface, a visual motor policy, and local adaptation. The semantic interface changes the active intention. The local controllers change how that intention is executed using current measurements.

### 4.1. Reusing a visual pose-conditioned student

A frozen CNN predicts the image coordinates and depth of the object center and eight keypoints from a $120 \times 120$ RGB image. Calibration, unprojection, and rigid fitting recover $\hat{p}_t \in \mathbb{R}^7$, comprising position and an XYZW quaternion. The fit residual accompanies the measurement. The current rendering uses the object's Gaussian representation with live black hand occlusion. It is not a full-scene image or a real-camera observation, and no ground-truth pose fallback is used at deployment.

The inherited student expects geometric goals. We replace its goal channels with a learned projection of $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains target-free proprioceptive features and history, while $u_t \in \mathbb{R}^{21}$ gives normalized joint-target proposals. Compatible encoders, recurrent weights, actor weights, and normalization statistics are initialized from the released policy. Neither target quaternion nor relative-goal quaternion is an input to the migrated student.

The first training stage fits the new interface while preserving inherited coefficients. A second stage adapts the recurrent core using actions labeled by the frozen privileged teacher. Teacher and student rollouts are mixed in a DAgger-style procedure [7]. The frozen release uses 200 interface updates and 400 adaptation updates, each with 128 environments and 45 rollout steps, totaling 3,456,000 transitions. These stages use simulated pose noise. All reported closed-loop trials instead obtain pose from the frozen CNN. We separately measure the loss introduced by this transfer rather than assuming that initialization preserves the teacher's performance.

### 4.2. Semantic grounding and public actions

We maintain six normalized goal vectors $B = \{g_1, \ldots, g_6\}$ from cached SigLIP2 text features [9]. Each vector denotes

a color skill and is bound to a canonical orientation in the audited asset. The goal bank is fixed for the cube release.

**Astra interface.** Astra receives the requested letter and visual observations. It emits a validated public action from `inspect_face`, `reorient_target`, `verify_target`, `stop`, or `abstain`. A color argument selects a vector from $B$. Inspection can change the motor goal without declaring that the requested letter has been found. Inspection actions can request between 10 and 90 control steps. Subsequent visual evidence and public action history support another decision. We use the name Astra for the hosted route `gpt-6-astra` recorded in our experiment, not for a model trained here. Its exact reproducible provider version is a release requirement.

The vector delivered to the student is a local encoding of a public command. We do not access Astra's internal features, train Astra weights, or describe this pathway as end-to-end language-to-action learning. This explicit interface also enables fixed-public-plan diagnostics in which the motor system is evaluated under recorded decisions and delays.

**Fine-tuned VLM.** The compact baseline uses Qwen3-VL-4B-Instruct [6] with low-rank adaptation and a goal head. The head predicts a normalized continuous vector in the centered five-dimensional span of $B$, together with a readability probability. The continuous vector enters Eq. 3 directly. Its nearest canonical vector determines the declared binding and the orientation used by the shared stopping gate. This interface is not unrestricted open-vocabulary motor generation.

Let $y$ indicate whether the requested mark is readable, $k$ its skill index, and $\hat{g}$ the predicted goal. Training combines readability binary cross-entropy with positive-example goal classification and cosine alignment:

$$\mathcal{L}_{\text{ground}} = \text{BCE}(\hat{y}, y) + \mathbf{1}[y = 1]\big(\text{CE}(\hat{k}, k) + 1 - \cos(\hat{g}, g_k)\big). \tag{4}$$

The implementation averages positive terms over readable examples. Readability supervision uses geometric visibility and an image-signal filter, not human judgments. Training first uses teacher-view data and then includes observations from student trajectories. The current objective does not include an action-consistency loss.

**Rule VLM.** The rule baseline parses a color and readability decision from the local VLM and maps accepted colors to $B$. Both local baselines use the same fixed inspection sequence when a view is not accepted. They use the migrated student directly, with no learned waiting actor, execution actor, or learned completion correction. This is the primary complete-system comparison. A separate auxiliary comparison adds the same control actors to each provider to examine the control contribution.

### 4.3. Learning control while a request is pending

An outstanding query does not determine whether the current movement is still safe. We construct a deployable feature vector $s_t$ from proprioception, measured pose, student belief, proposed action, pose-fit residual, semantic binding status, and request status. It also includes the measured pose and command at the request anchor and elapsed pending time. The active goal $g_t$ is encoded separately. Before a binding is established it denotes the startup or inspection skill, not the unavailable future target.

A 192-dimensional sensor encoder and 64-dimensional goal encoder feed a 128-dimensional fusion layer. A waiting actor produces one speed logit $r_t$ and a residual $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ be normalized current joint position and $\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \text{sigmoid}(4r_t + \text{logit}(0.15)), \tag{5}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{6}$$

$$a_t = \text{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{7}$$

Zero outputs reproduce fixed slowing with a coefficient of 0.15 during a pending query. The environment applies its original joint mapping and smoothing. The residual gives the controller a way to correct a holding posture rather than merely suppress movement.

Waiting targets are generated by the frozen teacher with the request anchor as the holding objective. This supervision is available in simulation. The deployed actor receives measurements and the active goal only. It never receives the answer being computed. The objective fits the applied action to the teacher label and penalizes residual magnitude:

$$\mathcal{L}_{\text{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{8}$$

### 4.4. Learning execution after a decision

A second actor with the same architecture serves non-pending execution. It uses the student proposal as its base, with no speed attenuation, and learns the bounded correction in Eq. 7. Its target is the teacher action for the active goal. It can remain active throughout execution, including inspection, rather than only during a short handover interval.

Paired training examples reuse a physical state and recurrent history with different valid goals. Features and teacher labels are reconstructed for each goal. The motivation is that a generic holding action can preserve contact without moving toward the requested orientation. Goal-dependent

labels give the actor a reason to retain directional information. This data construction does not itself demonstrate that the deployed network uses semantics. A matched no-goal actor, shuffled-goal evaluation, and a same-state/different-goal intervention are required to test that mechanism.

Training also includes states collected under recorded public action and delay schedules, matching the controller's pending and execution phases more closely to deployment. The selected execution actor is the v6 candidate from development. The same weights are retained throughout the frozen evaluation. The original development selection was informed by observed development outcomes, not a blind preregistration. Reserved evaluation results did not select the checkpoint. Learned completion heads were explored during development but are not active in the reported release.

### 4.5. Closed-loop execution

At each control step, the system updates visual pose, advances the recurrent student, chooses the waiting or execution actor, and applies the resulting joint targets. A completed semantic response is validated before it changes the active skill or declared binding. The next image reflects the motion that occurred while the previous response was pending. The common stopping gate is checked separately from semantic action generation. Invalid model output cannot supply ground-truth information or silently select the requested target.

This decomposition offers two distinct opportunities for learning. The low-level student can improve physical reach under its actual pose sensor. The controllers can improve how a fixed student handles a pending request or a new intention. These changes must be evaluated separately. Replacing a student changes the controller's belief and proposal distribution, so existing controller weights are checkpoint-bound and cannot be assumed to transfer unchanged.

## 5. Experiments

We organize evaluation around four questions: **Q1** How do the complete semantic systems compare? **Q2** Which local control components change outcomes? **Q3** Does improved visual pose prediction improve closed-loop success? **Q4** Do the findings transfer to a physical hand? The frozen cube study answers part of Q1 and Q2. Q3 and Q4 include explicit unfinished entries.

### 5.1. Setup and evidence scope

Experiments use the released Revo3 cube assets in an independent Isaac Lab 3.0/Newton environment. The validated scene has 23 bodies and 117 collision shapes. Contact capacities are fixed at 4,096 contacts and 1,024 constraints. Earlier runs with contact-buffer overflow are invalid and are excluded from this release. The original 320-trial suite contains 80 low-level diagnostics, 120 shared-control provider trials, and 120 additional Astra ablation trials. An 80-trial addendum evaluates the two bare VLM baselines. This gives 400 unique trials, including 320 semantic trials. Reusing an Astra cell across tables does not create new trials.

All semantic arms use the same scene, pose estimator, goal bank, student checkpoint, physical threshold, and shared stopping gate. The primary VLM baselines omit our learned control actors. For each arm, seeds 183101 and 183102 provide 20 trials each. Initial state matching is verified from recorded state values. Training, development, and evaluation seeds are separated, while repeated development measurements are not treated as held-out evidence. The bare-baseline addendum was declared after the earlier suite had run, following a clarification of baseline scope. It selected no new weights and changed no previous outcome.

### 5.2. Q1: Complete-system comparison

Table 1 reports the frozen cube results. Astra with learned control succeeds in 29 of 40 raw trials. The fine-tuned VLM succeeds in 27 and the rule VLM in 14. Excluding the shared service-affected trial gives 29/39, 26/39, and 13/39. The excluded trial was successful for both bare VLMs, which explains why their numerators also change. It would be incorrect to compare an eligible Astra rate against a raw VLM rate.

Astra and the fine-tuned VLM share 20 eligible successes. Astra has nine exclusive successes and the VLM six, giving a two-sided exact McNemar $p = 0.607$. Astra and the rule VLM share 11 successes, with 18 Astra-only and two rule-only successes, giving $p = 0.000402$. These are exploratory, unadjusted paired tests. The present evidence shows a large observed improvement over the rule baseline but does not establish an advantage over the fine-tuned model. Failure to resolve that difference is also not evidence of equivalence.

**Failure structure.** The ten Astra drops occur during unbound inspection. Nine occur with no model request pending and one during a pending request. None occurs before the first skill is delivered. The bare fine-tuned VLM has nine drops, two pose false stops, and two wrong-binding failures. Of its drops, five occur during unbound inspection, one after an incorrect binding, and three after a correct binding. The bare rule VLM has 15 drops, nine wrong bindings, and two horizon failures. These categories separate physical control from semantic decisions. The full method does not minimize raw drops among the three systems, even though it has the largest raw success count.

| System | Raw successes | Raw drops | Eligible successes | Eligible rate (%) | 95% Wilson interval |
|---|---|---|---|---|---|
| Rule VLM (bare) | 14/40 | 15 | 13/39 | 33.3 | [20.6, 49.0] |
| Fine-tuned VLM (bare) | 27/40 | 9 | 26/39 | 66.7 | [51.0, 79.4] |
| Astra + learned control (ours) | 29/40 | 10 | 29/39 | 74.4 | [58.9, 85.4] |

Table 1. **Q1: frozen cube comparison.** Each system has 40 raw trials. A service exclusion in one Astra trial removes the same trial from all three methods for the eligible comparison, leaving 39. The fine-tuned and rule VLM baselines use the shared student without learned waiting or execution actors. The raw success fractions are 72.5%, 67.5%, and 35.0% for Astra, fine-tuned VLM, and rule VLM, respectively. Intervals describe individual eligible rates, not paired effect sizes.

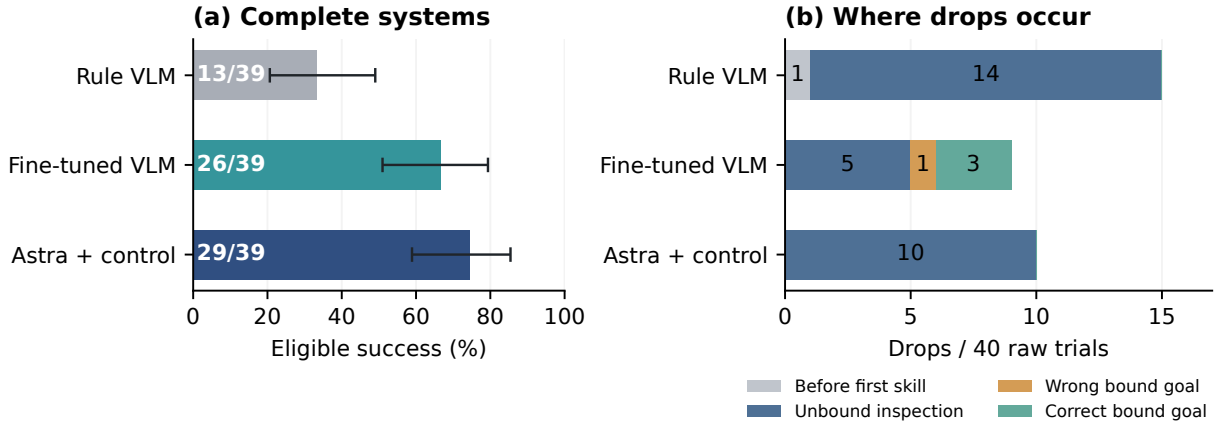



Figure 2. **Success and contact failure answer different questions.** Left: eligible success with 95% Wilson intervals on the common 39-trial panel. Right: all observed drops from 40 raw trials, classified by semantic execution context. Astra's ten drops all occur before a correct target is bound, during inspection. This concentration motivates improving evidence-seeking motion. It does not imply perfect control after binding, since reaching that phase selects a subset of trajectories.

**Model and control contributions.** As an auxiliary comparison, we retain the original runs that use the same learned controllers for every provider. Fine-tuned VLM with shared control succeeds in 30/40 raw trials and rule VLM in 16/40, with five and twelve drops. On that panel's common 39 eligible trials, Astra and fine-tuned VLM both succeed in 29, while rule VLM succeeds in 15. Thus an unconditional ranking of Astra above the fine-tuned VLM is not supported. The main comparison assesses complete systems with the requested baseline definitions. The auxiliary panel is needed to discuss the upper-level model independently of exclusive controller access.

**Inference and visual evidence.** The full Astra arm makes 94 model calls across its 40 raw trials. Recorded call latency has a median of 5.14 seconds and a 95th percentile of 7.15 seconds. One long call is service-excluded. These delays span many physical control steps. Usage is available for 93 calls and totals 154,859 reported tokens, so it is not a complete billing estimate. Figure 3 shows actual model inputs from the first formal Astra trial. Black areas include the background and hand occlusion. The example illustrates the observation domain, not a representative success-rate sample or a full-scene rendering.




Figure 3. Actual uncropped RGB requests from the first formal Astra episode, targeting T. The episode succeeds, but the displayed request images alone do not certify final orientation. Completion is measured independently by Eq. 1.

### 5.3. Q2: Waiting and execution ablations

Table 2 holds Astra and the student fixed and varies two control components. Fixed waiting uses coefficient 0.15. The execution baseline uses the unmodified student. Three service-affected trial identifiers are removed symmetrically for the four-way eligible comparison, leaving 37 trials per

| Waiting | Execution | Raw | Drops | Eligible | % |
|---|---|---|---|---|---|
| Fixed 0.15 | Student | 23/40 | 16 | 20/37 | 54.1 |
| Learned | Student | 26/40 | 11 | 25/37 | 67.6 |
| Fixed 0.15 | Learned | 28/40 | 8 | 26/37 | 70.3 |
| Learned | Learned | 29/40 | 10 | 27/37 | 73.0 |

Table 2. **Q2: Astra control ablation.** Raw successes and drops retain all 40 trials. Eligible successes use the common 37-trial set. The full cell is the same Astra release as Table 1, evaluated on a different shared exclusion set.

| Policy, all with frozen CNN feedback | Reach | Drops |
|---|---|---|
| Released pose-goal student | 39/40 | 1 |
| Frozen latent student | 38/40 | 2 |
| Latent + simulated-noise DAgger | 39/40 | 1 |
| Latent + CNN-feedback DAgger | 40/40 | 0 |

Table 3. **Q3: completed successor development diagnostic.** Each policy uses the same 40 fresh initial states. These 160 trials are separate from the earlier 39/40 versus 35/40 frozen diagnostic and from semantic success. All physical-log validity checks pass. Predictor improvement and real-robot results remain unmeasured.

arm.

Learned waiting alone increases eligible successes from 20 to 25. Learned execution with fixed waiting gives 26, and the full configuration gives 27. The full versus fixed-control paired comparison has $p = 0.210$. Full versus waiting-only gives $p = 0.791$, and full versus execution-only gives $p = 1.0$. These counts motivate the method but do not establish that both components are independently necessary or that their combination has a reliable interaction effect.

The execution-only arm has eight raw drops, fewer than the full method's ten, yet also has a pose false stop, wrong binding, and abstention. This illustrates why reducing drops alone does not ensure successful completion. In both fixed-waiting arms, four episodes drop before a first skill arrives. The full configuration survives that phase in all four matched cases, but only two subsequently succeed and two drop during later inspection. This is a useful phase-level observation, not a causal estimate from four examples.

The next mechanism study must separate learned semantics from added capacity. **[TODO: Fit a no-goal actor with the same state inputs, parameter budget, and data budget. Compare it with the full actor on matched trials. Add a same-state/different-goal intervention, shuffled-goal control, and an anchor-free or time-only actor.]** A separate closed-loop study must keep the Astra model and control actors fixed while comparing a single initial call, repeated queries without history, and the full feedback interface. These experiments test whether visual evidence and semantic intention change actions in a useful way.

### 5.4. Q3: Perception and student adaptation

The current low-level diagnostic obtains pose from the same frozen CNN for both policies. The released pose-goal student reaches its target before dropping in 39/40 trials, while the migrated latent student reaches in 35/40. Drops rise from one to five. Thirty-five trials succeed for both, four only for the released student, and one for neither. The exact paired $p = 0.125$ does not establish a reliable transfer gap, but the observed loss identifies a concrete development target. These results measure physical reach, not autonomous semantic completion, and must not be substituted into Table 1.

A separate successor development study has now completed. Two branches start from the same latent student and train for 400 additional updates with 16 environments and 45 steps per update. Each receives 288,000 new transitions. One uses simulated pose noise, the other actual frozen CNN predictions. The teacher, CNN, architecture, budget, and initialization remain fixed. On 40 fresh matched CNN-feedback trials, the released student reaches in 39, the frozen latent student in 38, extra simulated-noise training in 39, and extra CNN training in all 40, with zero drops (Table 3). This trains the student to use its sensor, not a better pose predictor. The one- and two-trial gains are not statistically resolved. The CNN candidate passes the development admission rule but is not deployed into the frozen semantic comparison.

**[TODO: Insert the improved-predictor versus original-predictor comparison and oracle-pose diagnostic.]** The planned predictor study reports translation and rotation errors with an independent reference, then tests closed-loop performance with the policy and agent held fixed. An oracle-pose diagnostic estimates remaining control limitations but is never a deployable baseline. A second, separately labeled stage may adapt the policy to the new predictor. Jointly changing predictor and policy in the first comparison would prevent attributing the gain to perception.

### 5.5. Q4: Additional objects and physical transfer

The present evidence concerns one marked cube. Additional object checkpoints will be integrated through the same semantic goal interface, with object-specific geometry, calibration, pose predictor, motor policy, and skill definitions recorded. An object-specific student is legitimate but does not demonstrate zero-shot object transfer. We plan approximately five objects with different grasp geometries and visual ambiguities rather than five texture variants of the same cube.

On hardware, the same three systems should share the physical hand, camera, reset procedure, instruction set, and stopping definition. We plan 40 trials per system and object, divided into two sessions of 20. Real resets need a docu-

| Object / domain | Rule | Fine-tuned | Full |
|---|---|---|---|
| Cube / simulation | 14/40 | 27/40 | 29/40 |
| Object 2 / simulation | TODO | TODO | TODO |
| Object 3 / simulation | TODO | TODO | TODO |
| Object 4 / simulation | TODO | TODO | TODO |
| Object 5 / simulation | TODO | TODO | TODO |
| Cube / physical hand | TODO | TODO | TODO |
| Other physical objects | TODO | TODO | TODO |

Table 4. **Planned breadth study, with only one completed row.** Entries retain raw denominators. Eligible comparisons and uncertainty will accompany every completed object. No macro-average is computed over unfinished rows.

mented matching tolerance rather than a claim of exact simulator state equality. An independent reference must measure the target orientation at stopping. Using the deployed predictor to both stop and certify success would conflate its bias with task performance. Episode videos, model request timing, joint commands, and termination events should be recorded together.

**[TODO: Add the physical setup photograph, object dimensions and masses, camera calibration, control frequency and latency, per-object success table, and representative failures.]** The main comparisons should first stay within the original scene distribution. Lighting and background variation are useful later studies, but they do not replace evidence that the semantic controller and physical policy solve the intended task.

## 6. Discussion and Limitations

The current results suggest that semantic inspection and contact-preserving execution must be evaluated together. Astra's remaining drops concentrate before target binding, so a controller trained only to hold during inference cannot solve the full problem. Evidence-seeking actions must also move safely through unbound states. The complete system has the highest observed raw success in the primary comparison, but it neither eliminates drops nor establishes superiority over a fine-tuned VLM. A stronger paper needs to explain which decisions benefit from a stronger agent and which failures are dominated by perception or motor control.

The fixed cube arrangement permits a shortcut from the requested letter to its color skill. Current success does not rule out memorizing that relation. A layout permutation, image masking or shuffling intervention, and paired alternative targets are necessary to substantiate visual semantic grounding. Such interventions can retain the original background and lighting. They test the claimed mechanism rather than broad appearance generalization.

The deployed visual domain is currently a Gaussian-rendered object with hand occlusion. Simulator capacity checks establish valid execution of this chosen model, not physical fidelity or real-world robustness. A proprietary semantic service also limits exact model replication. Its public prompt, tool schema, route version, sampling settings, outputs, and measured latencies must accompany a final experimental release. No safety guarantee, broad object generalization, or statistically resolved advantage over the fine-tuned baseline is claimed from this cube study.

## 7. Conclusion

We presented a pose-mediated semantic reorientation system with learned local control around asynchronous visual decisions. It preserves a common motor interface while comparing an external agent, a fine-tuned goal predictor, and a rule-based VLM. The frozen cube results establish an executable baseline and locate the main remaining failures in unbound inspection. The outstanding experiments are explicit: causal tests of semantic control and feedback, improved visual perception under fixed control, additional object policies, and independently scored real-hand trials. Together these studies can determine whether the proposed control interface provides a repeatable advantage for agentic dexterous manipulation.

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## LLM Usage Statement

Generative AI tools assisted with research discussion, implementation, experiment orchestration, literature retrieval, data analysis, and manuscript drafting. Astra and Qwen3-VL are also components evaluated in the research. Numerical results in this draft are derived from recorded experiments. Unfinished experiments are explicitly marked TODO. The authors must verify the final text, references, code, and results and remain responsible for the submitted work.